

SHASHANK KESAVAMATHAM

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EDUCATION

The Ohio State University - Bachelor of Science in Computer Science and Engineering August 2024 - Summer 2027

- **Concentration:** Cybersecurity and Artificial Intelligence
- **Related Coursework:** Information Security, Data Structures, Databases, Discrete Structures
- **Certificates:** CompTIA Security+ (Expected October 2026)
- **Clearance:** DoD Secret Clearance (Inactive)
- **Achievements:** 1st Place, BDAA Machine Learning Competition (CNN malware image classification model, **94% accuracy**, **50+ teams**)

SKILLS

- **Security & Compliance:** DoD STIG Hardening, OpenSCAP, Nessus, Vulnerability Assessment, Security Remediation
- **Cloud & Infrastructure:** AWS, Docker Containers
- **Identity & Access / Security Monitoring:** OpenLDAP, Kerberos, SSH Key Auth, Splunk Enterprise
- **Programming & Automation:** Python, Java, C, SQL, Bash

PROFESSIONAL EXPERIENCE

Griffiss Institute – Viceroy Envoy Internship (UTSA NSCC, supporting 16th Air Force / DoD) San Antonio, TX

Cybersecurity Intern - Automated Vulnerability Management Pipeline

June 2026 – August 2026

Skills: Nessus API | Vuln. Management | Splunk | RHEL 9 | Docker | SSH Auth. | STIG Compliance | LDAP/SSSD

- Designed and built a **Python** automation pipeline integrating the **Nessus REST API** with **Splunk HEC**, reverse-engineering undocumented API behavior to extract structured **JSON** vulnerability data (CVEs, CVSS, VPR, EPSS) from scan-status endpoints despite Nessus Essentials export limitations, reducing scan-to-dashboard time-to-visibility from 15 minutes to under 2 seconds (**99.81% reduction**) across an **8-device lab fleet**.
- Deployed and configured **Nessus Essentials** in **Docker** on a dedicated **Linux** scanner host, conducted authenticated vulnerability scans across the lab environment (**273 findings: 24 critical, 61 high-severity**), and developed a **Splunk** dashboard with custom **SPL** queries and visualizations for severity, exploitability, and host-level vulnerability analysis
- Performed **DoD STIG** remediation on **RHEL 9** systems using **OpenSCAP/SCAP Workbench**, independently reaching **99% compliance** and collaborating with team members to close remaining gaps, achieving **100% compliance** across user access controls, DNS persistence, SELinux, and hardening baselines.
- Implemented centralized **SSH key-based authentication** (RSA 4096-bit) via **LDAP/SSSD integration** across all lab hosts, replacing password-based access and reducing unauthorized-access attack surface.
- Extended the **Splunk** deployment with **Universal Forwarder**-based log collection (syslog/auth.log) across **7 of 8 devices**, cutting manual multi-device security log review from ~103 seconds to 3 seconds (**97% reduction**) via centralized search.

Triangle Cyber – AWS Cloud Security Internship

Remote

Cloud Security Intern - AWS Threat Simulation & Security Monitoring

May 2026 – August 2026

Skills: AWS EC2 | Splunk | CloudTrail | GuardDuty | Nmap | SQL Injection | XSS | VPC Networking | SIEM

- Simulated a **5-vector attack chain** (recon, brute-force, command injection, SQLi, XSS) against DVWA in AWS, achieving **100%** detection confirmation across **3** correlated Splunk log sources
- Built a **5-panel** Splunk dashboard, audited AWS CloudTrail for infrastructure changes, and triaged GuardDuty findings to distinguish genuine threats from background noise

Extern (Outamation)

Remote

AI Externship - AI-Powered Document Intelligence Pipeline

April 2026 – June 2026

Skills: RAG (LlamaIndex) | Claude API | OCR (EasyOCR/Tesseract) | Embeddings | Gradio | Python

- Architected and launched an end-to-end **RAG pipeline** (**LlamaIndex**, **Claude Haiku**, **BGE embeddings**) for mortgage document Q&A, achieving **100% accuracy** across test queries while correctly declining to answer when information was absent from the document.
- Benchmarked **3 OCR engines** (**Tesseract**, **EasyOCR**, **PaddleOCR**) on scanned mortgage docs (~17x **speed/accuracy trade-off**) and extracted **8** structured fields from digital PDFs via **PyMuPDF** bounding-box coordinates and regex, exceeding the **6-field requirement**.
- Developed a **Claude Haiku**-based document classification pipeline segmenting a **7-page** mixed-document PDF into **3 distinct types** with **7/7 pages correctly classified**, then shipped a **Gradio** chatbot to **Hugging Face Spaces** supporting hybrid digital/scanned PDF ingestion.

PROJECT EXPERIENCE

AES-Powered Steganography Messenger Tool

August 2025 – September 2025

Skills: Cryptography | Pillow | AES | Steganography

- Built a Python messaging tool combining **AES-GCM** encryption (**PBKDF2 key derivation**) with LSB-based image steganography, embedding encrypted messages with **<0.03%** pixel distortion for secure, visually-intact transmission.